

AHMED ISMAIL ABDUALLAH

Junior AI Engineer · Machine Learning & MLOps Specialist

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PROFESSIONAL SUMMARY

Results-driven Junior AI Engineer with hands-on expertise in designing and deploying end-to-end machine learning pipelines, deep learning architectures, and production-grade AI systems. Proficient in Computer Vision, Natural Language Processing with Attention Models, and Transfer Learning. Experienced in MLOps workflows including experiment tracking with MLflow, model lifecycle management via Hugging Face, and cloud deployment on Azure. Passionate about building scalable AI solutions that deliver measurable real-world impact.

EDUCATION

Bachelor of Technology in Computer Science

Aug 2022 – Jul 2026 (Expected)

Zagazig University, Faculty of Computer Science & Information — Specialization: Computer Science & Artificial Intelligence

Relevant Coursework & Professional Training

- Prompt Engineering for Large Language Models
- Mathematics for AI: Statistics & Linear Algebra
- Python Programming & Python for Data Science, AI & Development
- Data Preprocessing, Feature Engineering & Visualization
- Machine Learning Techniques (Supervised & Unsupervised)
- Deep Learning Architectures: CNNs, RNNs, and Modern Variants
- Natural Language Processing with Attention Models & Transformers
- Computer Vision & Transfer Learning
- MLOps Tools: MLflow, Hugging Face Hub & Model Deployment
- Azure AI Fundamentals · Azure AI Engineer Associate
- Capstone Project: End-to-End AI System Design & Deployment

PROFESSIONAL EXPERIENCE

Machine Learning Engineering Intern – Digital Egypt Pioneers Initiative (DEPI) Jan 2026 – Present

- Architected and implemented end-to-end AI pipelines spanning data ingestion, preprocessing, model training, evaluation, and cloud deployment.
- Developed production-ready deep learning models for Computer Vision (CNNs, Transfer Learning) and NLP (Transformers, Attention Mechanisms) using TensorFlow.
- Applied systematic hyperparameter tuning, regularization, and model optimization to achieve measurable performance gains.
- Managed experiment lifecycle and model versioning using MLflow and Hugging Face Hub, ensuring reproducibility and traceability across iterations.

Data Science Intern – almentor

Jun 2025 – Dec 2025

- Conducted comprehensive exploratory data analysis (EDA) and built predictive ML models across structured datasets.

- ▶ Engineered domain-specific features and applied advanced statistical analysis to improve model accuracy and generalization.
- ▶ Delivered executive-ready insights through interactive data visualizations and structured analytical reports.

KEY PROJECTS

Tree Clinic — AI-Powered Agricultural Assistant | *Zagazig University* Aug 2025 – Present

- ▶ Built a smart mobile application that detects tree diseases from leaf images using deep learning (CNN + Transfer Learning).
- ▶ Integrated organic and chemical treatment recommendations with an in-app marketplace, and deployed an intelligent agricultural chatbot to assist farmers.
- ▶ Technologies: Python · TensorFlow · Flutter · Computer Vision · Transfer Learning

Exo AI — Exoplanet Classification Platform | *Zagazig University* Nov 2025

- ▶ Developed an AI-powered platform automating exoplanet classification from NASA Kepler, K2, and TESS datasets using ensemble ML models.
- ▶ Integrated Python ML backend with a .NET 9 REST API and Flutter web frontend, delivering a full-stack end-to-end AI solution.
- ▶ Technologies: Python · Scikit-learn · .NET 9 · Flutter · REST API

Movie Intelligence System — Sentiment & Genre Analysis Jul 2025

- ▶ Designed an intelligent analytics system for movies using NLP-based sentiment analysis, genre classification, and audience engagement modeling.
- ▶ Extracted actionable insights from audience feedback to support content strategy and rating trend forecasting.
- ▶ Technologies: Python · NLP · Transformers · Data Science · Pandas · Matplotlib

TECHNICAL SKILLS

Languages	Python · SQL
ML / AI	Supervised & Unsupervised Learning · Model Evaluation · Hyperparameter Tuning
Deep Learning	CNNs · RNNs · Transformers · Attention Models · Transfer Learning (TensorFlow)
NLP	Text Classification · Sentiment Analysis · Sequence Modeling · Hugging Face
Computer Vision	Image Classification · Object Detection · Feature Extraction · OpenCV
Data Science	NumPy · Pandas · Matplotlib · Scikit-learn · EDA · Feature Engineering
MLOps	MLflow · Hugging Face Hub · Model Deployment · Monitoring · Experiment Tracking
Cloud	Azure AI Fundamentals · Azure AI Engineer Associate
Prompt Eng.	LLM Prompting · Chain-of-Thought · Few-Shot Learning · Instruction Tuning